Lab 6: Demand Experimentation and Pricing Optimization

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DS 5030 Understanding Uncertainty
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Lab 6: Demand Experimentation and Pricing Optimization

Finding the revenue-maximizing price is difficult to compute when the demand curve is unknown. This experiment simulates seller chosen prices between $0 and $70 to learn about demand by watching how randomly arriving customers respond. Each tested price also results in a cost, as customers shown a poor price produce less revenue. The goal is to find an optimal price with a target revenue of $11.925 per customer using as few experimental customers as possible.

Method

Experimental Design and Uncertainty

The model handles uncertainty by comparing prices on mean revenue per arriving customer and only discarding a price when a two-sample Z-test shows it to be significantly worse than the current leader. Because the number of arriving customers is random and does not depend on price, revenue per customer is a fair comparison: it removes noise of slow or busy days and leaves only the effect of price on purchase decisions. For each tested price, the algorithm records the totals of customers, revenue, and squared revenue, resulting in the mean and standard error, allowing earlier data to be reused when a price is tested again.

To find possible prices, every candidate price is compared to the best price with the gap test below, using a one-sided cutoff of $Z \geq 1.645$. A price stays if its gap from the leading price is smaller than 1.645 standard errors. The 90% level was chosen over 95% or 99% because wrongly keeping a near-optimal price costs little, while demanding stronger evidence would require far more customers at a worse price.

$$Z = \frac{\bar{R}_{best} - \bar{R}_{p}}{\sqrt{SE^2_{best} + SE^2_{p}}} \; ; \; drop\ p\ if\ Z \geq 1.645$$

Search Strategy

Gradient ascent strategy was attempted first by estimating the slope of revenue from 300 customers on each side of the current price and steps uphill, but it only climbs towards the nearest peak. Because the revenue curve has several peaks, the result depends on the starting price and can settle on a suboptimal peak depending on where it starts. At estimates close to the peak, the curve is too flat and the price wanders instead of settling on an answer. For these reasons, it was not selected.

The second and final approach searches globally, then locally. A coarse grid of 14 prices ($5 to $70, in $5 steps) was tested with batches of 100 customers. After each round, prices significantly worse than the leader are excluded and the candidates are tested again for up to 10 rounds. The same pruning technique was then repeated on a $1 grid from $5 below and above the coarse winner. Finally, the $1 grid was re-run with 30 random seeds to check for stability.

Coarse Grid With Z-Test Pruning Results

The two stage grid search selected $28.00 using 5,730 customers and earning $53,309 during the experiments.

Stage	Prices tested	Price selected	Customers	Revenue	Share of optimal
Coarse grid	5-70, $5 steps	$30	4,638	$42,330	76.6%
Finer grid	25-35, $1 steps	$28	1,092	$10,979	84.4%
Total		$28	5,730	$53,309	78.0%

The coarse stage uses most of the customers because it starts with prices far from the peak before the pruning removes them. The finer stage earned $10.05 per customer because every price tested was near the peak.

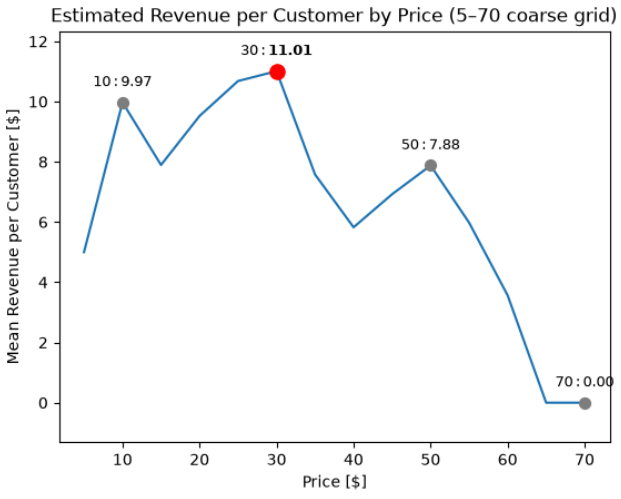


Figure 1. Revenue per batch at each coarse-grid price ($5 to $70 in $5 steps). The line shows the average batch revenue and the shaded band shows its 95% confidence interval. Revenue peaks between $25 and $30, so the search moved to that region.

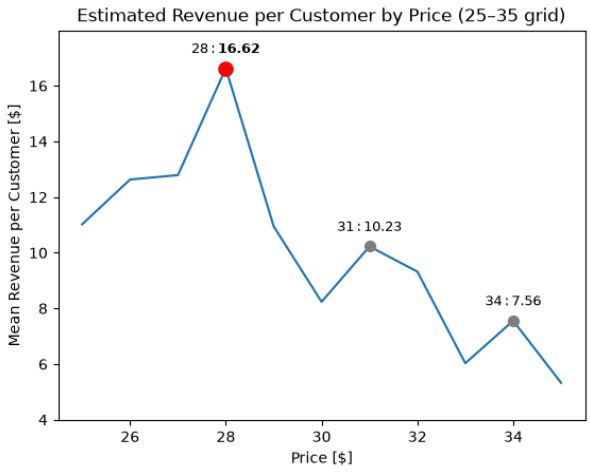


Figure 2. Estimated mean revenue per customer on the $1 grid from $25 to $35. The highest estimate is at $28, the selected price. Prices dropped early by the Z-test have small samples, which is why the curve is jagged.

Across 30 seeds, the finer grid stage chose $28 in 13 runs, $27 in 9, $26 in 5, $29 in 2, and $25 once. It used 4,555 customers on average (range 1,995-6,421) and captured 92.0% of optimal revenue on average (SD 2.6, range 85.3% to 97.9%). Every selected price was within $3 of $28, and 29 of 30 were within $2.

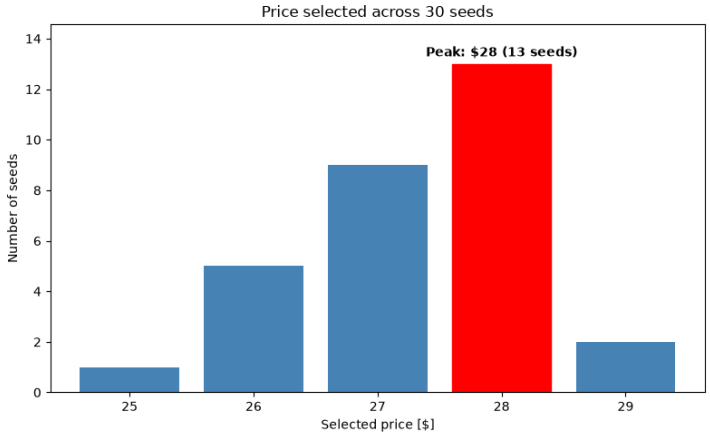


Figure 3. Price selected by the $1-grid stage across 30 random seeds. $28 was chosen most often (13 runs), followed by $27 (9 runs), and every run chose a price between $25 and $29.

Discussion

Challenges and Resolutions

The first challenge was that the revenue curve has several peaks, so gradient ascent starting at $45 or $59 got stuck near $52, earning only ~64% of optimal. The final search resolved this by first sampling the full $5 to $70 range with a coarse grid, then refining on a $1 grid around the winner.

The second challenge was the trade off between certainty and cost. Near the optimum, prices such as $27 and $28 differ by only a few cents per customer, so telling them apart takes thousands of customers. This was resolved with small batches of ~100 customers, a 90% cutoff, and pooling each price's data across rounds, while accepting near-optimal prices such as $26 or $27, which captures ~92% of optimal revenue on average across 30 seeds.

Conclusion

The two-stage search, a coarse grid with Z-test pruning followed by a $1 grid, selected a price of $28 while experimenting on 5,730 customers and earning $53,309 during the experiments. This is 78% of the optimal $11.925 per customer across the entire search, 84.4% within the finer stage alone, and 92% across 30 seeds of the finer stage on average. Searching the full range first and refining kept the algorithm from being trapped on a local peak.